SQL PURCHASE BEHAVIOUR ANALYSIS

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DATASET: **Predict Customer Purchase Behaviour Dataset (Kaggle)**

VARIABLES:

- **Age:** Customer's age
- **Gender:** Customer's gender (0: Male, 1: Female)
- **Annual Income:** Annual income of the customer in dollars
- **Number of Purchases:** Total number of purchases made by the customer
- **Product Category:** Category of the purchased product (0: Electronics, 1: Clothing, 2: Home Goods, 3: Beauty, 4: Sports)
- **Time Spent on Website:** Time spent by the customer on the website in minutes
- **Loyalty Program:** Whether the customer is a member of the loyalty program (0: No, 1: Yes)
- **Discounts Availed:** Number of discounts availed by the customer (range: 0-5)

QUERY 1: Identifying which age demographic transacts the most

Result Grid | Filter Rows:

SUM(NumberofPurchases)	Age category
3100	36-49
2274	50-60
2088	61-70
1863	18-25
1754	26-35

- Thus, we can see that the maximum number of purchases have been done by the age category of **36-49,** beating out the second most (50-60) by almost 1000 more purchases
- We can infer that the key demographic making purchases in this store is in the middle-aged group, followed by senior citizens, whereas the younger 2 categories (18-25 and 26-35) make the least number of purchases.

QUERY 2: In the 36-49 age category, which product categories are the most popular

Product Category	COUNT(ProductCategory)
Sports	70
Electronics	59
Clothing	59
Beauty	53
Home Goods	45

- In the age group that makes the most purchases (36-49), we can then analyse which product categories are the most popular among them. We find that to be sports and electronics, with the least number of purchases for home goods.
- A possible inference we can make here is as individuals settle into their jobs, and attain a steady flow of income, their expenditure shifts towards recreational goods, whereas daily necessities (demand for which is relatively inelastic) consumes less of their income

Query 3: Comparing Number of purchases within a category to average purchases in that category, using window function

Query 4: Identifying top spenders in each category using CTE and dense_rank() window function

ProductCategory	Age	AnnualIncome	NumberOfPurchases	purchase_rank
0	40	98472.83607271877	20	1
0	40	36167.89187006633	20	1
0	50	109555.08926651179	20	1
0	37	70632.64633731465	20	1
0	50	92821.86385003946	20	1
0	52	77639.40186929441	20	1
0	70	85814.58773437636	19	2
0	24	98084.63865128627	19	2
0	46	50866.35088872482	19	2
0	48	122216.1659199243	19	2
0	45	126073.10206592025	19	2
0	38	62927.95843329468	19	2
0	49	138198.62676200282	19	2
0	67	102703.05741207025	19	2
0	18	46262.0368923237	19	2

- We can identify the characteristics of individuals who make the most purchases in a respective category
- This helps us formulate targeted advertisements or sales strategies

Query 5: Checking correlation between time spent on site and no of purchases, to determine how strong the intent to make a purchase was when logging into the site